

4.3.49

EE25BTECH11059 - Vaishnavi Ramkrishna Anantheertha

Question: Write the equation of the lines for which $\tan \theta = \frac{1}{2}$, where θ is the inclination of the line, and

- (a) y intercept $-\frac{3}{2}$
- (b) x intercept 4

Solution 1:

Variable	Value
A	$(0, -\frac{3}{2})$
m	$\frac{1}{2}$

TABLE 0: Variables Used

Let $\mathbf{B}$ be a point on the line

$$\mathbf{A} = \begin{pmatrix} 0 \\ -\frac{3}{2} \end{pmatrix} \quad (0.1)$$

$$\text{Let } \mathbf{M} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (0.2)$$

$$\mathbf{M} = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} \quad (0.3)$$

$$(\mathbf{B} - \mathbf{A}) = \mathbf{M} \quad (0.4)$$

$$\mathbf{B} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (0.5)$$

Let eq of line be

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.6)$$

As $\mathbf{A}$ and $\mathbf{B}$ lie on the line

$$\mathbf{n}^T \mathbf{A} = 1 \quad (0.7)$$

$$\mathbf{n}^T \mathbf{B} = 1 \quad (0.8)$$

From eq (0.7) and (0.8)

$$\begin{pmatrix} 0 & -\frac{3}{2} \\ 1 & -1 \end{pmatrix} n = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.9)$$

In augmented matrix form

$$\left[\begin{array}{cc|c} 0 & -\frac{3}{2} & 1 \\ 1 & -1 & 1 \end{array} \right] \quad (0.10)$$

Swap R_1 and R_2

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & -\frac{3}{2} & 1 \end{array} \right] \quad (0.11)$$

$$R_2 \rightarrow -\frac{2R_2}{3}$$

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & -\frac{2}{3} \end{array} \right] \quad (0.12)$$

$$R_1 \rightarrow R_1 + R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \end{array} \right] \quad (0.13)$$

hence eq of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.14)$$

$$\left(\frac{1}{3} \quad -\frac{2}{3} \right) \mathbf{x} = 1 \quad (0.15)$$

Refer to Figure

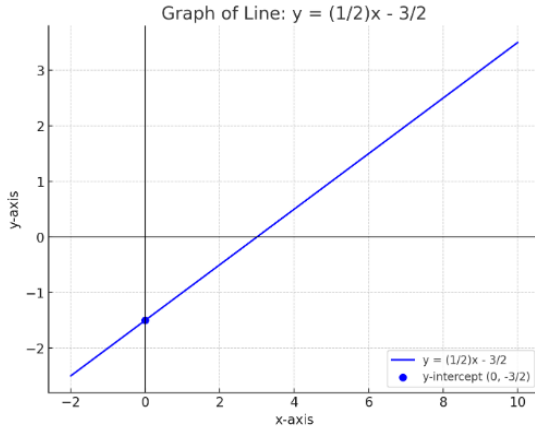


Fig. 0.1

Solution 2:

Variable	Value
A	$(4, 0)$
m	$\frac{1}{2}$

TABLE 0: Variables Used

Let $\mathbf{B}$ be a point on the line

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad (0.16)$$

$$\text{Let } \mathbf{M} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (0.17)$$

$$\mathbf{M} = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} \quad (0.18)$$

$$(\mathbf{B} - \mathbf{A}) = \mathbf{M} \quad (0.19)$$

$$\mathbf{B} = \begin{pmatrix} 5 \\ \frac{1}{2} \end{pmatrix} \quad (0.20)$$

Let eq of line be

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.21)$$

As $\mathbf{A}$ and $\mathbf{B}$ lie on the line

$$\mathbf{n}^T \mathbf{A} = 1 \quad (0.22)$$

$$\mathbf{n}^T \mathbf{B} = 1 \quad (0.23)$$

From eq (0.22) and (0.23)

$$\begin{pmatrix} 4 & 0 \\ 5 & \frac{1}{2} \end{pmatrix} \mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.24)$$

In augmented matrix form

$$\left[\begin{array}{cc|c} 4 & 0 & 1 \\ 5 & \frac{1}{2} & 1 \end{array} \right] \quad (0.25)$$

$$R_1 \rightarrow \frac{R_1}{4}$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 5 & \frac{1}{2} & 1 \end{array} \right] \quad (0.26)$$

$$R_2 \rightarrow R_2 - 5R_1$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & -\frac{1}{4} \end{array} \right] \quad (0.27)$$

$$R_2 \rightarrow 2R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \end{array} \right] \quad (0.28)$$

$$\mathbf{n} = \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{2} \end{pmatrix} \quad (0.29)$$

hence eq of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.30)$$

$$\begin{pmatrix} \frac{1}{4} & -\frac{1}{2} \end{pmatrix} \mathbf{x} = 1 \quad (0.31)$$

Refer to Figure

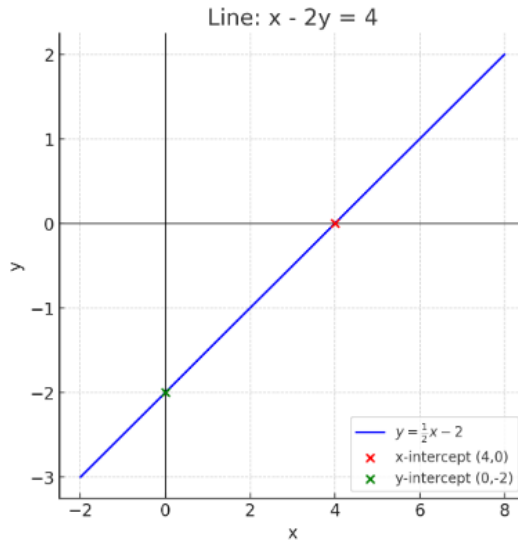


Fig. 0.2